

Nihar Masurkar

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Summary

Robotics Ph.D. candidate building autonomous inspection systems that combine robotic manipulation with acoustic, ultrasonic, and infrared thermographic nondestructive evaluation (NDE). Hands-on across the full stack: sensor and hardware integration, force-feedback control, signal processing, simulation, and field data collection. 6 peer-reviewed journal articles, 11 conference papers, and a U.S. provisional patent, with work that translates biological sensing strategies into deployable robotic inspection. Focused on bringing contact acoustic, ultrasonic, and thermal inspection to mobile robotic platforms.

Education

Ph.D. in Robotics and Automation Engineering

2023 - Present

Specialization: Non-Destructive Testing

GPA: 3.87

Arizona State University, Mesa, Arizona

Master of Science in Engineering, Robotics & Autonomous Systems

2019 - 2022

Concentration: Mechanical & Aerospace

GPA: 3.46

Arizona State University, Tempe, Arizona

Academic Projects & Research

Autonomous Robotic Ultrasonic Inspection, ISNDE Lab, ASU

Jan 2023 – Present

- Built an automated ultrasonic testing (UT) cell integrating a UR5e manipulator (RTDE), Sonatest Wave flaw detector, ZED stereo camera, and an ESP32-driven couplant pump over MQTT, operated through a custom PyQt5 console with live A-scan display, jogging, and device health monitoring.
- Designed a PID force controller for probe coupling and quantified A-scan repeatability against direct-force and threshold-descent strategies, showing force feedback as a real-time measure of coupling quality (CEECT 2026).
- Integrated contact and air-coupled ultrasonic transducers on manipulators and mobile platforms, using fiducial markers to register scans to a pipeline digital twin and motion capture to validate joint trajectories.
- Co-developed RoboScan, an autonomous UT thickness-measurement framework (ASNT 2026), and deep learning time-series classification of non-contact UT signals for pipe condition assessment (ASME JNDE 2023).

Bio-Inspired Tap-Glide Acoustic Inspection Robot, ISNDE Lab, ASU

Jan 2023 – Present

- Conducted multimodal behavioural studies with aye-ayes (*Daubentonia madagascariensis*) at the Duke Lemur Center, synchronizing high-speed infrared thermography (FLIR X6800sc, 250 fps) with acoustic recording across six subjects.
- Identified frictional gliding as a foraging mode complementary to percussive tapping, supported by kinematic and acoustic evidence (Scientific Reports, under review; Int. J. Primatology 2025; SPIE 2024; QNDE 2024).
- Modelled the third digit as a 2R manipulator (Denavit-Hartenberg parameters, forward and inverse kinematics) from motion tracking data (Biomimetic Intelligence and Robotics 2023).
- Designed and built a cam-driven test bench reproducing tapping (8 Hz and above) and gliding with machined aluminium cams, closed-loop DYNAMIXEL actuation, FUTEK load cell force sensing, and a 5x-scaled finger (IEEE ICRA 2027, under review).
- Translated the newly identified aye-aye foraging motion (frictional gliding) into an acoustic tap-testing analysis pipeline for composite inspection, targeting sub-surface flaws and delamination; filed a related U.S. provisional patent.

Acoustic Gas Leak Localization on Robotic Platforms, ISNDE Lab, ASU

Jan 2024 – Present

- Developed an automated robotic approach that localizes minor gas leaks using a two-microphone array mounted on a robotic platform (ASNT 2024).
- Co-developed hearing-guided motion planning for precise leak localization (Materials Evaluation 2025; ASNT 2025) and bio-inspired active acoustic sensing with robotic pinnae (ICCAS 2026).
- Contributed to field evaluation of an optical microphone against conventional detectors for woolly bat echolocation recording in Malaysia (Scientific Reports, under review).
- Contributed to biomimetic studies of termite drumming as a model for vibro-acoustic sensing (IEEE ROBIO 2024).

TurtleBot 4 Object Detection and Path Following, Academic Project, ASU

Course Project

- Set up a TurtleBot 4 (Create 3 base, Raspberry Pi) on ROS 2 Humble and Ubuntu 22.04, including DDS middleware and multi-machine network discovery.
- Developed ROS 2 Python nodes that detect objects in the onboard camera stream and drive the robot along a target path using closed-loop feedback.

Foldable Five-Bar Bipedal Walking Robot, Academic Project, ASU

Course Project

- Designed an ostrich-inspired bipedal walker using foldable laminate five-bar leg mechanisms; performed kinematic analysis and built a parametric MuJoCo model for static and dynamic gait simulation.

- Optimized end-effector link length for distance travelled, prototyped cardstock and fiberglass laminate versions from DXF cut files with Arduino-controlled servos and validated measured end-effector trajectories against simulation.

Octopus-Inspired Robotic Gripper, BIRTH Lab, ASU

Jan 2021 – May 2021

- Designed a modular soft pneumatic gripper inspired by octopus tentacles and elephant trunks, built to grasp objects securely with minimal contact area.
- Developed a pneumatic controller for silicone arms with central air channels. In testing, a two-finger 200 mm gripper held 45 N in maximum pull-off and achieved full 360-degree curling.
- Extended the work to intermittently actuated, bio-inspired suction cups for contact-based robotic inspection (IEEE ICRA 2027, under review).

Lizard Inspired Tube Inspection Robot, BIRTH Lab, ASU

Jan 2020 – Dec 2023

- Developed a PID-controlled three-axis Cartesian test rig with load cell feedback to characterize gripper pad friction for a robot that grips the outside of a pipe and translates along its length (Int. J. Intelligent Robotics and Applications 2024).
- Fabricated and optimized PDMS friction pads critical to the robot's friction-based mobility, and designed a closed-loop, encoder-based controller for precise gripper opening, closing, speed, load torque, and position.
- Designed the EMAT mounting and gripper integration for a modular robotic gripper with electromagnetic acoustic transducers, enabling couplant-free ultrasonic inspection of pipes and tubular components during robotic traversal (Materials Evaluation 2021; ASNT Research Symposium 2021).

Patents

- N. Masurkar** and E. Dehghan-Niri, “Systems and Methods for Bio-Inspired Rub and Friction Mechanisms for Acoustic and Vibro-Acoustic Non-destructive Testing and Evaluation”, U.S. Provisional Patent Application 63/875,171, filed Sep. 3, 2025.

Publications

- N. Masurkar**, H. Nemati, and E. Dehghan-Niri, “Aye-Aye Percussive Foraging Integrates Frictional Gliding and Percussive Tapping: Kinematic and Acoustic Evidence”, *Scientific Reports, Under Review*, 2026.
- E. Dehghan-Niri, M. Nasr, **N. Masurkar**, M. A. I. Azmi, N. I. M. Baharom, M. D. H. M. Khairi, V. L. Sien, A. H. M. Azmi, F. A. A. Khan, and R. Müller, “Listening Beyond Limits: High-Resolution Insights into Woolly Bat (*Kerivoula hardwickii*) Calls via Next-Generation Wildlife Acoustic Sensing”, *Scientific Reports, Under Review*, 2026.
- H. Nemati, **N. Masurkar**, and E. Dehghan-Niri, “Investigating Acoustic-Based Foraging Behavior in Aye-Ayes (*Daubentonia madagascariensis*) Through Infrared Thermography”, *International Journal of Primatology*, vol. 46, no. 3, pp. 763–784, 2025, doi: [10.1007/s10764-025-00498-y](https://doi.org/10.1007/s10764-025-00498-y).
- M. Nasr, **N. Masurkar**, P. Meshkizadeh, and E. Dehghan-Niri, “Guided by Hearing: Robotic Motion Planning for Precision in Gas Leakage Localization”, *Materials Evaluation*, vol. 83, no. 4, pp. 51–61, 2025, doi: [10.32548/2025.me-04491](https://doi.org/10.32548/2025.me-04491).
- N. Masurkar**, A. Das, M. Rudraboina, D. Morris-Sjolund, F. Alvidrez, E. Dehghan-Niri, and H. Marvi, “A Biologically-Inspired Tube Inspection Robot with Friction-Based Mobility”, *International Journal of Intelligent Robotics and Applications*, vol. 9, no. 1, pp. 164–175, 2024, doi: [10.1007/s41315-024-00370-x](https://doi.org/10.1007/s41315-024-00370-x).
- N. Masurkar**, K. Jiming, H. Nemati, and E. Dehghan-Niri, “Aye-Aye Middle Finger Kinematic Modeling and Motion Tracking During Tap-Scanning”, *Biomimetic Intelligence and Robotics*, vol. 3, no. 4, Art. no. 100134, 2023, doi: [10.1016/j.birob.2023.100134](https://doi.org/10.1016/j.birob.2023.100134).
- S. Hespeler, H. Nemati, **N. Masurkar**, F. Alvidrez, H. Marvi, and E. Dehghan-Niri, “Deep Learning-Based Time-Series Classification for Robotic Inspection of Pipe Condition Using Non-Contact Ultrasonic Testing”, *Journal of Nondestructive Evaluation, Diagnostics and Prognostics of Engineering Systems*, vol. 7, no. 1, 2023, doi: [10.1115/1.4063694](https://doi.org/10.1115/1.4063694).
- H. Nemati, F. Alvidrez, A. Das, **N. Masurkar**, M. Rudraboina, H. Marvi, and E. Dehghan-Niri, “Integrating Electromagnetic Acoustic Transducers in a Modular Robotic Gripper for Inspecting Tubular Components”, *Materials Evaluation*, vol. 79, no. 7, pp. 715–727, 2021, doi: [10.32548/2021.me-04223](https://doi.org/10.32548/2021.me-04223).

Conferences

- N. Masurkar** and E. Dehghan-Niri, “Bio-Inspired Tap-Glide Robotic Testing”, 2027 IEEE International Conference on Robotics and Automation (ICRA), *Under Review*.
- V. Kurakalva, H. Marvi, E. Dehghan-Niri, N. Layos, A. Kabir, and **N. Masurkar**, “Intermittently Actuated, Bio-Inspired Suction Cups for Contact-Based Robotic Inspection”, 2027 IEEE International Conference on Robotics and Automation (ICRA), *Under Review*.
- N. Masurkar**, G. Sandys, and E. Dehghan-Niri, “Automated Ultrasonic Testing Using a Robotic Manipulator with Force-Feedback Assessment of Probe Coupling Quality”, 2026 8th International Conference on Electrical Engineering and Control Technologies (CEEET), Bangkok, Thailand, Dec. 17–20, 2026.
- E. Dehghan-Niri, M. Nasr, H. Nemati, **N. Masurkar**, and M. Haghighesnas-Jaryani, “Laying the Foundations for Bioinspired Active Acoustic Sensing in Gas Leak Localization”, 2026 26th International Conference on Control, Automation and Systems (ICCAS), Sapporo, Japan, 2026, accepted.

5. G. Sandys, **N. Masurkar**, and E. Dehghan-Niri, “RoboScan: A Robotic Framework for Autonomous Ultrasonic Testing for Thickness Measurement”, 2026 ASNT Research Symposium, Salt Lake City, UT, USA, 2026.
6. H. Nemati, **N. Masurkar**, and E. Dehghan-Niri, “Aye-Aye Inspired Acoustic Sensing for Robotic Tap-Testing: From Auditory Morphology to Bioinspired Sensor Design”, 2025 9th International Conference on Automation, Control and Robotics (ICACR), Xi'an, China, Nov. 28–30, 2025.
7. M. Nasr, **N. Masurkar**, P. Meshkizadeh, and E. Dehghan-Niri, “Gas Leakage Localization Through Bio-Inspired Motion Strategy on Robotic Platforms”, 2025 ASNT Research Symposium, Indianapolis, IN, USA, Jun. 23–27, 2025.
8. E. Dehghan-Niri, **N. Masurkar**, H. Nemati, Z. Goode, H. S. Halliday, and J. Liebig, “Overcoming Challenges in the Biomimetic Study of Termite Drumming Behavior”, 2024 IEEE International Conference on Robotics and Biomimetics (ROBIO), Bangkok, Thailand, 2024, pp. 172–177, doi: [10.1109/ROBIO64047.2024.10907719](https://doi.org/10.1109/ROBIO64047.2024.10907719).
9. **N. Masurkar**, H. Nemati, and E. Dehghan-Niri, “Advancing the Biomimetic Design for Automated Tap Scanning Through Aye-Aye's Behavioral Studies Utilizing High-Resolution Thermal Cameras”, 51st Annual Review of Progress in Quantitative Nondestructive Evaluation (QNDE), Denver, CO, USA, Jul. 22–24, 2024.
10. **N. Masurkar**, P. Meshkizadeh, and E. Dehghan-Niri, “Automated Robotic Approach for Source Localization of Minor Gas Leak Using a Two-Microphone Array”, 32nd ASNT Research Symposium, Pittsburgh, PA, USA, Jun. 24–28, 2024.
11. **N. Masurkar**, H. Nemati, and E. Dehghan-Niri, “Enhancing Biomimetic Design of Tap Scanning Sensors Through High-Resolution Thermal Camera-Based Behavioral Studies”, in Proc. SPIE, vol. 12944, Bioinspiration, Biomimetics, and Bioreplication XIV, May 9, 2024, Art. no. 129440E, doi: [10.1117/12.3027268](https://doi.org/10.1117/12.3027268).
12. **N. Masurkar**, H. Nemati, E. Dehghan-Niri, and S. Hespeler, “An Automated Robotic Approach for Inspection of Metallic Tubular Structure”, 31st ASNT Research Symposium, Columbus, OH, USA, Jun. 26–30, 2023, doi: [10.32548/rs.2023.003](https://doi.org/10.32548/rs.2023.003).
13. H. Nemati, F. Alvidrez, A. Das, **N. Masurkar**, M. Rudraboina, H. Marvi, and E. Dehghan-Niri, “Toward Automated Ultrasonic Inspection of Pipelines and Tubular Components”, ASNT Research Symposium, Virtual, Mar. 2021.

Technical Skills

- **Robotics & Control:** ROS 2 Humble, UR5e (RTDE), DYNAMIXEL servos, PID and force-feedback control, kinematics (DH, FK/IK), motion capture, MuJoCo.
- **NDE & Sensing:** Ultrasonic testing (contact, air-coupled, EMAT), Sonatest Wave, acoustic tap testing, microphone arrays, infrared thermography (FLIR X6800sc), ZED stereo vision, Ouster LiDAR, load cells
- **Signal Processing & Simulation:** MATLAB (FFT, CWT, time-frequency analysis).
- **Programming & Software:** Python, C++, PyQt5, MQTT, YOLO object detection and tracking.
- **Embedded & Electronics:** ESP32, Arduino, PCB design (KiCAD, Autodesk Eagle, ExpressPCB)
- **CAD & Fabrication:** Autodesk Fusion 360, SolidWorks, 3D printing, foldable laminate fabrication, PDMS and silicone molding, precision machined mechanisms
- **Cloud & IoT:** AWS EC2, AWS IoT Core, Amazon Kinesis Video Streams

Professional Experience

IoT Developer, University Technology Office, ASU

Aug 2020 – May 2023

- Developed a point-cloud clustering and machine learning classification pipeline for Ouster OS0-32 LiDAR data to estimate pedestrian density and monitor vehicle traffic around campus security poles.
- Designed a multi-sensor environmental monitoring circuit that streams data to AWS IoT Core over MQTT.
- Deployed PoE IP cameras streaming live video through Amazon Kinesis Video Streams, and built an optimized YOLO-based multiple object tracking (MOT) script to detect and classify people around security poles.

Awards

- 1st place, Devils Invent: Drone Defense (ASU hackathon sponsored by the Department of Homeland Security and Boeing), as part of Team KCAaN. Designed a system to protect critical buildings from aerial threats.
- 1st place, Starbucks Hackathon. Designed an augmented reality display that lets baristas view order details hands-free.
- 2nd place, Devils Invent: Protecting America's Public Access Areas (30-hour hackathon), as part of Team BAHMMN. Developed an early-response system that uses dynamic exit signage to disperse crowds efficiently.
- Research funding and sponsorship. Secured ASU Engineering MORE program funding and W.L. Gore & Associates sponsorship for the Lizard-Inspired Tube Inspection Robot.

Leadership & Service

- Session Chair, 51st Annual Review of Progress in Quantitative Nondestructive Evaluation (QNDE), Denver, CO, July 2024.
- Mentored 4 undergraduate and 5 graduate researchers over five years in robotics, NDE, and experimental methods.
- K-12 robotics and NDE outreach (lab tours, demos, school visits) for ASU's School of Manufacturing Systems and Networks.